



June 26, 2025

Kansas Department of Health & Environment
Bureau of Air
1000 SW Jackson, Suite 310
Topeka, Kansas 66612-1366

**RE: Hugoton Compressor Station
Stevens County, Kansas
Source ID: 1890015
Notification of Compliance Status for U-164**

To Whom It May Concern:

In accordance with 40 CFR 63.6645(h), Merit Energy Company, LLC (Merit) is submitting this Notification of Compliance Status following the completion of a performance test for one Waukesha L7042 GSI 1,232 horsepower compressor engine, U-164, at the Hugoton Compressor Station (Facility). The semi-annual performance test was completed on May 15, 2025. The information required by 40 CFR 63.9(h)(2)(i) is listed below.

- A. The methods that were used to determine compliance:
The methods that were used to determine compliance are detailed in Section 3.0 of the attached performance test report.
- B. The results of any performance tests, opacity or visible emission observations, continuous monitoring system (CMS) performance evaluations, and/or other monitoring procedures or methods that were conducted:
The results of the performance test are attached.
- C. The methods that will be used for determining continuing compliance, including a description of monitoring and reporting requirements and test methods:
Continuous compliance will be determined by monitoring the catalyst inlet temperature and monthly monitoring of the pressure drop across the catalyst in accordance with Table 6 of 40 CFR 63, Subpart ZZZZ. Semi-annual performance tests will also be completed.
- D. The type and quantity of hazardous air pollutants emitted by the source (or surrogate pollutants if specified in the relevant standard), reported in units and averaging times and in accordance with the test methods specified in the relevant standard:
U-164 emits formaldehyde. See the results of the attached performance test.

- E. If the relevant standard applies to both major and area sources, an analysis demonstrating whether the affected source is a major source (using the emissions data generated for this notification):

Hugoton Compressor Station has a potential to emit greater than 25 tons per year (TPY) for total hazardous air pollutants (HAPs) and greater than 10 TPY for a single HAP; therefore, Hugoton Compressor Station is a major source.

- F. A description of the air pollution control equipment (or method) for each emission point, including each control device (or method) for each hazardous air pollutant and the control efficiency (percent) for each control device (or method):

U-164 is equipped with a non-selective catalyst which limits the formaldehyde to 350 ppbvd or less at 15 percent oxygen. The catalyst specification sheet is attached.

- G. A statement by the owner or operator of the affected existing, new, or reconstructed source as to whether the source has complied with the relevant standard or other requirements:

The statement that the affected source has complied with 40 CFR 63, Subpart ZZZZ is attached.

- H. A written report of the average percent load determination:

In accordance with 40 CFR 63.6620(i), the average percent load determination is included in the attached performance test report.

If you have any questions regarding this notification or require additional information, please contact me at bailey.sharp@meritenergy.com or at (972) 628-1454.

Sincerely,

MERIT ENERGY COMPANY



Bailey Sharp
Environmental Analyst

Certification of Truth, Accuracy and Completeness

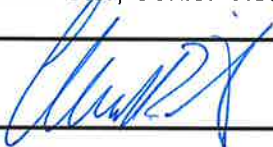
40 CFR 63.9(h)(2)(i)(G)

Based on information and belief formed after reasonable inquiry, I certify that the statements and information contained in this notification of compliance status are true, accurate, and complete to the best of my knowledge and U-164 at Hugoton Compressor Station has complied with the applicable requirements of 40 CFR 63, Subpart ZZZZ.

Name

(typed): Chad Brister, Senior Vice President

(Signed):



Date:

6/26/25

Equipment Specification

Proposal Information

Proposal Number: RJM-22-007577 Date: 12/7/2022
Project Reference: Waukesha 7042

Engine Information

Engine Make: Waukesha
Engine Model: 7042
Rated Speed: 1000 RPM
Fuel Description: Natural Gas
Hours Of Operation: 8760 Hours per year
Load: 100%
Speed: 1800 RPM
Power Output: 896 bhp
Exhaust Flow Rate: 5,153 acfm (cfm)
Exhaust Temperature: 1,060 ° F
O₂: 0.3%
H₂O: 18.5%

Emission Data (100% Load)

Emission	Raw Engine Emissions						Target Outlet Emissions						Calculated Reduction
	g/bhp-hr	tons/yr	ppmvd @ 15% O ₂	ppmvd	g/kW-hr	lb/MW-hr	g/bhp-hr	tons/yr	ppmvd @ 15% O ₂	ppmvd	g/kW-hr	lb/MW-hr	
NO _x *	13	112.48	704	2,456	17.433	38.43	4	34.61	216	756	5.364	11.83	69.2%
CO	9	77.87	800	2,793	12.069	26.61	4	34.61	356	1,241	5.364	11.83	55.6%

System Specifications

Catalyst (Replacement Catalyst)

Element Model Number: MECB-TW-RO-2650-0000-350
Number of Catalyst Layers: 1
Number of Catalyst Per Layer: 1
Catalyst Back Pressure: 3.0 inWC (Clean)
Design Exhaust Flow Rate: 5,153 acfm
Design Exhaust Temperature: 1,060f
Dimensions: Ø 26.5 in
Exhaust Temperature Limits**: 750f – 1250f (catalyst inlet); 1350f (catalyst outlet)
System Pressure Loss: 3.0 inWC (Clean)

* MW referenced as NO₂

** General catalyst temperature operating range. Performance is based on the Design Exhaust Temperature.



303 West 3rd Street • Elk City, Oklahoma 73644 • 580-225-0403

May 28, 2025

Kansas Department of Health and Environment
1000 SW Jackson Street
Topeka, KS 66612

RE: Merit Energy Company
40 CFR Part 63 Subpart ZZZZ – Test Report Submittal

To Whom It May Concern:

We are notifying you of a performance testing report for 40 CFR Part 63 Subpart ZZZZ on behalf of Merit Energy. Please see below for the information pertinent to the testing. Testing took place the day of May 15, 2025.

Facility/Unit	Unit ID	Serial Number	Location
Hugoton Compressor Station	APC-164	PP168	Stevens County, Kansas

For information regarding the performance test, you can contact me by email at klackey@gasinc.us or by phone at 580-225-0403. For information regarding engine or location information, please contact Bailey Sharp by email at bailey.sharp@meritenergy.com or by phone at 972-628-1454.

Regards,

Katie Lackey

Katie Lackey
Client Services Development Supervisor
GAS Inc.

Index

1.0 Key Personnel.....	3
2.0 Sampling System.....	3
3.0 Methods Used.....	3
4.0 Test Summaries.....	5
5.0 Run Summaries.....	6
6.0 Volumetric Flow Rate Data.....	7
7.0 Calculations.....	8
8.0 Oxygen Calibration.....	9
9.0 Engine Parameter Data Sheet.....	10
10.0 QA/QC Results.....	11
11.0 D6348 Annexes.....	12
12.0 Signature Page.....	18
13.0 Appendices.....	19
14.0 Bottle Certs.....	20
15.0 Tri Probe Certification via GD-031.....	26
16.0 GAS ALT 141_FTIR EPA.....	27
17.0 Tester Qualifications (resume).....	29
18.0 Raw Data.....	30

Tables

Table 5.1 (Run Summaries).....	6
Table 6.1 (Volumetric Flow Rate Data).....	7
Table 6.2 (Stack Gas Measurements).....	7
Table 8.1 (Oxygen Calibration).....	9
Annex Table 1.2.1 (Certified Calibration Bottle Concentrations).....	12
Annex Table 1.2.2 (Measurement System Capabilities).....	12
Annex Table 1.3.1 (Test Specific Target Analytes).....	13
Annex Table 4.1 (Measure System Capabilities).....	15

Figures

Figure 6.1 (Location of Traverse Points per Method 1).....	7
Annex Figure 1.4.1 (Sampling Train).....	13
Annex Figure 1.4.2 (Sampling Points).....	14
Annex Figure 1.4.3 (Sampling Port Locations).....	14

Appendices

Certified Calibration Bottle Certificates.....	20
Tri Probe Certification via GD-031.....	26
GAS ALT 141_FTIR EPA.....	27
Tester Qualifications (resume).....	29
Raw Data.....	30

1.0 Key Personnel

GAS
Merit Energy Company

Hunter Newton
Stacy Burrows

2.0 Sampling System

The sampling system used consisted of a Stainless steel probe, heated Teflon line, gas conditioning system, and a Gasmeter model DX4000 FTIR analyzer. The gas conditioning system used was a Gasmeter Personal Sampling System with a Zirconium Oxide oxygen sensor.

3.0 Methods Used

ASTM D6348-03

This extractive FTIR based field test method is used to quantify gas phase concentrations of multiple target analytes (CO, NOX, CH₂O, & VOC's) from stationary source effluent. Because an FTIR analyzer is potentially capable of analyzing hundreds of compounds, this test method is not analyte or source specific. The analytes, detection levels, and data quality objectives are expected to change for any particular testing situation. It is the responsibility of the tester to define the target analytes, the associated detection limits for those analytes in the particular source effluent, and the required data quality objectives for each specific test program. Provisions are included in this test method that require the tester to determine critical sampling system and instrument operational parameters, and for the conduct of QA/QC procedures. Testers following this test method will generate data that will allow an independent observer to verify the valid collection, identification, and quantification of the subject target analytes.

EPA Method 1 & 1A

The purpose of the method is to provide guidance for the selection of sampling ports and traverse points at which sampling for air pollutants will be performed pursuant to regulations set forth in this part.

EPA Method 2 & 2C

This method is applicable for the determination of the average velocity and the volumetric flow rate of a gas stream. The average gas velocity in a stack is determined from the gas density and from measurement of the average velocity head with a standard pitot tube. Velocity readings are taken from each stack at 16 separate traverse points (Table 6.1) and used to determine the engines mass emissions rate, calculated utilizing the formulas seen in section 7.0 of this report.

EPA Method 3A

This is a procedure for measuring oxygen (O₂) and carbon dioxide (CO₂) in stationary source emissions using a continuous instrumental analyzer. Quality assurance and quality control requirements are included to assure that the tester collects data of known quality. Documentation to these specific requirements for equipment, supplies, sample collection and analysis, calculations, and data analysis will be included.

Unit APC-164 with a serial number of PP168 which is a Waukesha L7042GSI engine located at Hugoton Compressor Station and operated by Merit Energy Company was tested for emissions of: Formaldehyde. The test was conducted on 5/15/2025 by Hunter Newton with Great Plains Analytical Services, Inc. All quality assurance and quality control tests were within acceptable tolerances.

The engine is a natural gas fired Rich Burn (4 Cycle) engine rated at 1232 brake horse power (BHP) at 1000 RPM. The engine was operating at 1107 BHP and 898 RPM which is 89.83% of maximum engine load during the test. The test HP calculation can be found on page 8. The engine was running at the maximum load available at the test site.

This test will satisfy the testing requirements for 40 CFR Part 63 Subpart ZZZZ. Unit APC-164 is authorized to operate under permit #1890015.

Site Verification Photos



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4.0 Test Summary

5

Engine/Compressor Specs					
Location	Hugoton Compressor Station	Unit ID	APC-164		
Make	Waukesha	Site Elevation ft.	3100		
Model	L7042GSI	Atmospheric Pressure psi.	13.15		
Serial Number	PP168	Stack Diameter in.	12.5		
mfg. rated hp	1232	Catalyst	Yes		
mfg. rated rpm	1000	Date of Manufacture	1/1/1975		
Engine/Compressor Operation		Run 1	Run 2	Run 3	3 Run Averages
Test Horsepower		1122	1099	1099	1107
Test RPM		911	892	892	898
Percent Load %		91.10%	89.20%	89.20%	89.83%
Intake Manifold Pressure (hg)		47.00	46.00	48.00	47.00
Intake Manifold Temperature (F)		157.00	155.00	150.00	154.00
Ambient Conditions					
Ambient Temperature Dry (F)		68.00	74.00	76.00	72.67
Exhaust Flow Data					
Q Stack (dscfh)		72785.85	72249.10	72147.26	72394.07
Q Stack (dscm/hr)		2061.05	2045.85	2042.96	2049.95
Moisture Fraction Bws		0.18	0.18	0.18	0.18
Results					
Method 3A Corrected O2% Dry		0.00%	0.00%	0.00%	0.00%
Moisture %		18.28%	18.16%	18.11%	18.18%

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Permitted Standards				Results			
	Federal	State	Local	10:40 AM	11:56 AM	01:07 PM	5/15/2025
			ZZZZ	Run 1	Run 2	Run 3	3 Run Average
CH2O (ppmvd) @15% O2				0.001	0.000	0.000	0.000
CH2O (ppbvd) @15% O2			350.000	0.691	0.282	0.000	0.324
CH2O (ppmvw)				0.002	0.001	0.000	0.001
CH2O (ppmvd)				0.002	0.001	0.000	0.001
CH2O (mol wt) 30.03							
Pass Permits							

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5.0 Run Summaries

Table 5.1 Run Summaries

Summary Source Run 1								
PPM Wet						CH ₂ O	Oxygen %	Moisture %
Baseline						0.03	0.00%	0.00%
Source						0.00	0.00%	18.28%
Spike						0.00	0.00%	16.30%
Source						0.00	0.00%	18.29%
Baseline						0.04	0.00%	0.00%
Avg. Wet						0.00	0.00%	18.28%
Summary Source Run 2								
PPM Wet						CH ₂ O	Oxygen %	Moisture %
Baseline						0.04	0.00%	0.00%
Source						0.00	0.00%	18.17%
Spike						0.00	0.00%	16.67%
Source						0.00	0.00%	18.15%
Baseline						0.03	0.00%	0.02%
Avg. Wet						0.00	0.00%	18.16%
Summary Source Run 3								
PPM Wet						CH ₂ O	Oxygen %	Moisture %
Baseline						0.03	0.00%	0.02%
Source						0.00	0.00%	18.11%
Spike						0.00	0.00%	16.99%
Source						0.00	0.00%	18.11%
Baseline						0.06	0.00%	0.00%
Avg. Wet						0.00	0.00%	18.11%

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Table 6.1. Data used for volumetric flow rate (Method 2)

Volumetric Flow Rate Data					
Pitot Tube Coefficient Cp(std)=	.99				
Stack diameter =	12.5	or:	1.04 feet	or:	.85 Square feet
			Run 1	Run 2	Run 3
			Average		
H2O	%		18.28	18.16	18.11
CO2	%d		11.31	11.42	11.24
O2	%d		0.00	0.00	0.00
CO	%d		0.44	0.46	0.46
Molecular Weight Stack Gas dry basis (Md)	g/g mole		29.81	29.96	29.93
Molecular Weight Stack Gas wet basis (Ms)	g/g mole		27.65	27.79	27.77
Stack Static Pressure (Pg)	"H2O		0.45	0.45	0.49
Stack Static Pressure (Pg)	"Hg		0.03	0.03	0.04
Atmospheric Pressure at Location (Pbar)	MBAR		1010.16	1006.44	1009.48
Atmospheric Pressure at Location (Pbar)	"Hg		29.84	29.73	29.82
Absolute Stack Pressure (Ps)	"Hg		29.87	29.76	29.86
Stack Temperature	Deg C		441.22	445.14	448.99
Stack Temperature	Deg F		826.19	833.25	840.19
Stack Temperature	Deg R		1285.86	1292.92	1299.86
Stack Gas Velocity	ft/sec		70.84	71.28	73.89
Stack Flow Rate Q	cfs		60.36	60.37	60.37
Stack Gas Wet Volumetric Flow Rate	scf/hr		89070.63	88278.76	88102.65
Stack Gas Dry Volumetric Flow Rate	scf/hr		72785.85	72249.10	72147.26
			72394.07		
Emissions Sampling Points - 3 point long line sampling probe					
Inches					
First Sampling Point Taken @ 16.7% of Stack Diameter					
2.09					
Second Sampling Point Taken @ 50% of Stack Diameter					
6.25					
Third Sampling Point Taken @ 83.3% of Stack Diameter					
10.41					

Volumetric Flow Rates

Table 6.2. Stack gas pressure readings measured with a standard pitot tube use for Volumetric Flow Rate

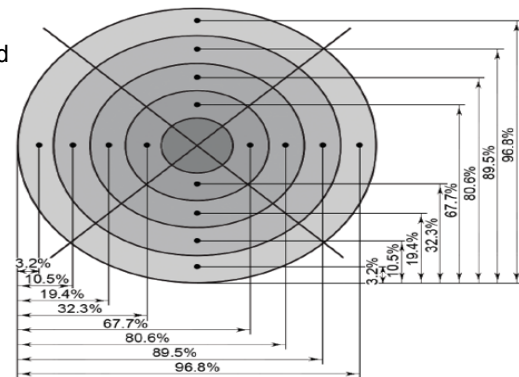
Δp_{std} = Velocity head measured by the standard pitot tube, (in.) H2O.

Pitot Tube Sampling Points (Velocity)	Run 1		Run 2		Run 3	
	Δp_{std} (in) H2O	Exhaust Temp F	Δp_{std} (in) H2O	Exhaust Temp F	Δp_{std} (in) H2O	Exhaust Temp F
1	0.31	826.00	0.61	833.00	0.37	840.00
2	0.30	827.00	0.55	831.00	0.44	841.00
3	0.28	828.00	0.53	835.00	0.48	840.00
4	0.33	824.00	0.51	833.00	0.41	838.00
5	0.36	825.00	0.39	832.00	0.52	832.00
6	0.38	826.00	0.33	834.00	0.48	839.00
7	0.41	824.00	0.48	833.00	0.60	841.00
8	0.48	827.00	0.46	835.00	0.61	844.00
9	0.55	828.00	0.49	836.00	0.62	845.00
10	0.63	827.00	0.51	833.00	0.57	832.00
11	0.67	825.00	0.47	832.00	0.55	844.00
12	0.60	826.00	0.44	833.00	0.51	845.00
13	0.55	828.00	0.43	831.00	0.48	841.00
14	0.50	827.00	0.38	835.00	0.43	842.00
15	0.42	826.00	0.36	832.00	0.38	840.00
16	0.36	825.00	0.32	834.00	0.33	839.00
Δp_{std} Average=	0.45	826.19	0.45	833.25	0.49	840.19

Sample after Back Purge:	0.33
Within 5% of Last Δp_{std} reading:	Yes
Stack Diameter (inches)	12.50
Inches upstream from disturbance	10.00
Inches downstream from disturbance	23.00

Figure 6.1

16 Traverse Points Were Used



Pitot readings are taken for Method 2 calculations using measuring points outlined in Method 1

*The exhaust stack did not present cyclonic flow conditions at the sampling location due to the absence of cyclones, inertial demisters, venturi scrubbers, or tangential inlets.

* Cyclonic Flow Check (Pass/Fail): PASS

Strat Check	
Check 1	279.737
Check 2	279.111
Check 3	283.953
Gas Used	methane
Pass/Fail	Pass

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7.0 Calculations

Method 2: Determination of Stack Gas Velocity and Volumetric Flow Rate

*Note- Use of this method negates the need for any fuel related numbers for emissions calculations

Nomenclature

$\Delta p(\text{avg})$ = Velocity head of stack gas, mm H₂O (in. H₂O).

3600 = Conversion Factor, sec/hr.

A = Cross-sectional area of stack, m² (ft²).

Bws = Water vapor in the gas stream (from ASTM D6348)

Cp(std) = Standard pitot tube coefficient; use 0.99

Kp = Velocity equation constant.

Md = Molecular weight of stack gas, dry basis, g/g-mole (lb./lb.-mole).

Ms = Molecular weight of stack gas, wet basis, g/g-mole (lb./lb.-mole).

Ps = Absolute stack pressure (Pbar+ Pg), mm Hg (in Hg)

Pstd = Standard absolute pressure, 760 mm Hg (29.92 in. Hg).

Qsd = Dry volumetric stack gas flow rate corrected to standard conditions, dscm/hr. (dscf/hr.).

Ts(abs) = Absolute stack temperature, °K (°R). = 460 + Ts for English units.

Tstd = Standard absolute temperature, 293°K (528 °R).

Vs = Average stack gas velocity, m/sec (ft./sec).

*(Observed from Method 3 12.3) Dry Molecular Weight. Equation 3-1

$$Md = .44(\%CO_2) + .32(\%O_2) + .28(\%N_2 + \%CO)$$

$$Md = .44(11.31) + .32(0) + .28(88.245 + .445) = 29.81 \text{ LB/LB-MOLE}$$

12.5 Molecular Weight of Stack Gas. Equation 2-6

$$Ms = Md(1 - Bws) + 18.0(Bws)$$

$$Ms = 29.81(1 - .18283) + 18.0(.18283) = 27.65 \text{ LB/LB-MOLE}$$

12.6 Average Stack Gas Velocity. Equation 2-7

$$Vs = Kp \cdot Cp(std) \cdot \sqrt{\Delta p_{avg}} \cdot \sqrt{(Ts(abs))/(Ps \cdot Ms)}$$

$$Vs = 85.49 \cdot .99 \cdot \sqrt{1285.86/(29.87 \cdot 27.65)} = 70.84 \text{ FT/SEC}$$

12.7 Average Stack Gas Volumetric Flow Rate. Equation 2-8

$$Qsd = 3600(1 - Bws)Vs \cdot A \cdot (Tstd \cdot Ps)/(Ts(abs) \cdot Pstd)$$

$$Qsd = 3600(1 - .18283)70.84 \cdot .852((528 \cdot 29.87)/(1285.86 \cdot 29.92)) = 72785.85 \text{ DSCF/HR}$$

*Standard conversion from feet to meters

$$Q = Qsd/35.315$$

$$Q = 72785.847/35.315 = 2061.046 \text{ DSCM/HR}$$

Emission Rates (Examples use CO Run 1)

Nomenclature

453.6 = Conversion factor lb. to gram

A = Cross-sectional area of stack, m² (ft²).

BHP/HR. = Brake work of the engine, horsepower-hour (HP-HR.).

BTU/HP-HR. = Brake Specific Fuel Consumption (HHV)

ER = Emission rate of (CO) in g/HP-hr.

F(d) = Volumes of combustion components per unit of heat content, scm/J (scf/million Btu).

Q = Stack gas volumetric flow rate, in standard cubic meters per hour, dry basis

LB/HR. = Emission rate of (CO) in LB/HR.

Mol wt. = Mol Weight of CO (28.01)

HP = Engines rated Horsepower

Mfg. = Manufacturer Exhaust flow rate at 100% (ft³/min)

O₂ = Concentration of oxygen on a dry basis, percent.

ppm = Parts Per Million (CO)

ppm@ 15% O₂ = PPM corrected to 15% O₂

Qsd = Dry volumetric stack gas flow rate corrected to standard conditions, dscm/hr. (DSCF/HR.).

Run Time = Run Time in Minutes

TPY = Tons per Year

Vs = Average stack gas velocity, m/sec (ft./sec).

CO corrected @ 15% O₂

$$\text{ppm @ 15\% O}_2 = \text{PPM} \cdot ((20.9 - 15\%O_2)/(20.9 - 0\%))$$

$$\text{ppm @ 15\% O}_2 = 4445.606 \cdot ((20.9 - 15)/(20.9 - 0)) = 1254.98 \text{ PPM @ 15\% O}_2$$

G/HP HR from 40 CFR Part 60 Subpart JJJJ

$$g/hp-hr = (\text{PPM} \cdot (1.164 \cdot 10^{-3}) \cdot Q \cdot (\text{Run Time}/60))/\text{BHP/HR}$$

$$(4445.606 \cdot (1.164 \cdot 10^{-3}) \cdot 2061.046 \cdot (\text{Run Time}/60))/1122.352 = 9.503 \text{ G/HP-HR}$$

LB/HR from Grams

$$\text{lb/hr} = \text{ER} \cdot 1/453.6 \cdot \text{BHP-HR}$$

$$\text{LB/HR} = 9.503 \cdot 1/453.6 \cdot 1122.352 = 23.512 \text{ LB/HR}$$

TPY

$$\text{TPY} = \text{LB/HR.} \cdot 4.38$$

$$\text{TPY} = 23.512 \cdot 4.38 = 102.983 \text{ TPY}$$

Horsepower

HP Provided On Site

Wet to Dry PPM

$$\text{ppm wet} \cdot (1/(1 - H_2O)) = \text{ppm dry}$$

$$4445.61 = 3632.8155 \cdot (1/(1 - .18))$$

8.0 Oxygen Calibration

8.1 Calibration error test; how do I confirm my analyzer calibration is correct? After the tester has assembled, prepared and calibrated the sampling system and analyzer, they conduct a 3-point analyzer calibration error test before the first run and again after any failed system bias test or failed drift test. They then introduce the low-, mid-, and high-level calibration gases sequentially in direct calibration mode. At each calibration gas level (low, mid, and high) the calibration error must be within ± 2.0 percent of the calibration span.

8.2 Initial system bias and system calibration error checks. Before sampling begins, it is determined whether the high- level or mid-level calibration gas best approximates the emissions and it is used as the upscale gas. The upscale gas is introduced at the probe upstream of all sample-conditioning components in system calibration mode.

(1) Next, zero gas is introduced as described above. The response must be within 0.5 percent of the upscale gas concentration.

(2) Low-level gas reading is observed until it has reached a final, stable value and the results are recorded. The measurement system will be operated at the normal sampling rate during all system bias checks.

(3) If the initial system bias specification is not met, corrective action is taken. The applicable calibration error test from Section 8.2.3 of EPA Method 7E is repeated along with the initial system bias check until acceptable results are achieved, after which sampling will begin. The pre- and post-run system bias must be within ± 5.0 percent of the calibration span for the low-level and upscale calibration gases.

8.3 Post-run system bias check and drift assessment - confirming that each sample collected is valid. Sampling may be performed for multiple runs before performing the post-run bias or system calibration error check provided this test is passed at the conclusion of the group of runs. A failed final test in this case will invalidate all runs subsequent to the last passed test.

(1) If the post-run system bias check is not passed, then the run is invalid. The problem is then diagnosed and fixed, then another calibration error test and system bias is passed before repeating the run.

(2) After each run, the low-level and upscale drift is calculated, using Equation 7E–4 in Section 12.5 from EPA Method 7E. If the post-run low- and upscale bias checks are passed, but the low-or upscale drift exceeds the specification in Section 13.3, the run data are valid, but a 3-point calibration error test and a system bias check must be performed and passed prior to additional testing.

Table 8.1 Oxygen Calibration

Method 7E 3.4 To the extent practicable, the measured emissions should be between 20 to 100 percent of the selected calibration span. This may not be practicable in some cases of low concentration measurements or testing for compliance with an emission limit when emissions are substantially less than the limit.						
EPA Method 3A QA Worksheet						
Certified Gas Concentraion Low-Level (%)		Certified Gas Concentraion Mid-Level (%)		Certified Gas Concentraion High-Level (%)		
0.00%		10.59%		20.92%		
(DIRECT) Analyzer Calibration Error (≤ 2%)					7E 8.5: Note: that you may risk sampling for multiple runs before performing the post-run bias provided you pass this test at the conclusion of the group of runs	
Linearity Check						
	Certified Concentration Value (%)	Direct Calibration Response (%)	Absolute Difference (%)	Analyzer Calibration Error (%)		
Zero Gas %	0.00%	0.00%	0.00%	0.00%		
Mid-Level Gas %	10.59%	10.52%	0.07%	0.33%		
High-Level Gas %	20.92%	20.80%	0.12%	0.59%		
(SYSTEM) Calibration Bias Checks (≤ 5%) and Drift Checks (≤ 3%)					Upscale Gas	10.59%
Zero Offset	0.00%	Bias Pre Initial Value		Bias Post Inital Values		
Span	20.92					
Analyzer Calibration Response (%)		System Calibrations Response Pre (%)	System Bias (% of Span) Pre	System Calibration Response Post (%)	System Bias (% of Span) Post	Drift (% of Span)
		0.00%	0.00%	0.00%	0.00%	0.00%
		Upscale Gas	10.52%	10.70%	0.86%	10.52%
(SYSTEM) Calibration Bias Checks (≤ 5%) and Drift Checks (≤ 3%)						
Avg. Gas Concentration (Run 1)		0.00%	Effluent Gas (Cgas) Run 1		0.00%	
Avg. Gas Concentration (Run 2)		0.00%	Effluent Gas (Cgas) Run 2		0.00%	
Avg. Gas Concentration (Run 3)		0.00%	Effluent Gas (Cgas) Run 3		0.00%	

EPA Method 3A QA Worksheet	
Zero Gas	100% Nitrogen
Mid-Level Gas	10.59%
High-Level Gas	20.92%

O2 QA/QC		
	Analyzer Direct Calibration Response	Certified bottle value %
Zero Gas %	0.00%	0.00%
Mid-Level Gas %	10.52%	10.59%
High-Level Gas %	20.80%	20.92%
System Calibration Response Pre (%)		
Zero Gas %	0.00%	Upscale Used
Upscale Cal	10.70%	10.59%
System Calibration Response Post (%)		
Zero Gas %	0.00%	Upscale Used
Upscale Cal	10.52%	10.59%

9.0 Engine Parameter Data Sheet



Company	Merit Energy Company
Facility	Hugoton Compressor Station
Date	5/15/2025
Site Elevation (ft)	3100
Unit ID	APC-164
Make	Waukesha
Model	L7042GSI
Serial Number	PP168
Technician	Hunter Newton

	Run 1	Run 2	Run 3	Completed
Run Start Times	10:40 AM	11:56 AM	01:07 PM	02:40 PM
Engine Hours	18815	18816	18817	18818

Engine Parameter Data				
	Run 1	Run 2	Run 3	Average
Engine Speed (RPM)	911.0	892.0	892.0	898.3
Intake Manifold Pressure (psi)	47.0	46.0	48.0	47.0
Intake Manifold Temp °F	157.0	155.0	150.0	154.0
Engine Load (BHP)	1122.4	1098.9	1098.9	1106.7
Ambient Temp °F	68.0	74.0	76.0	72.7
Humidity %	26.0	18.0	17.0	20.3
Dew Point °F	32.0	27.0	27.0	28.7
AFR Manufacturer/Type	Altronic	Altronic	Altronic	Altronic
Suction Pressure (psi)	27.0	27.0	27.0	27.0
Discharge Pressure (psi)	183.0	360.0	360.0	301.0
Catalyst (Yes or No)	Yes			
Catalyst Manufacturer	Miratech	Miratech	Miratech	Miratech
# of Catalyst Installed	1	1	1	1
Catalyst Inlet Temp °F	867.0	828.0	828.0	841.0
Catalyst Outlet Temp °F	812.0	811.0	811.0	811.3
Catalyst Pressure Drop H2O	0.0	0.0	0.0	0.0

10.0 QA/QC Results

Pre-test Baseline Results		Time:	Q A / Q C R e s u l t s
Carbon monoxide CO	Average: 0.05		
Nitrogen monoxide NO	Average: 0.10		
Nitrogen dioxide NO ₂	Average: 0.02		
NO _x	Average: 0.12		
VOC	Average: 0.03		
Oxygen	Average: 0.00		
CTS Direct to Analyzer		Time:	
CTS Bottle Concentration	Value: 99.57		
CTS Compound Concentration Avg	Value: 101.09		
Tolerance	2.00%		
Difference between measured and expected	1.52%		
Pre-test System Zero		Time:	
Carbon monoxide CO	Average: 0.00		
Nitrogen monoxide NO	Average: 0.00		
Nitrogen dioxide NO ₂	Average: 0.08		
NO _x	Average: 0.08		
VOC	Average: 0.06		
Oxygen	Average: 0.00		
Pre-test System CTS & Mechanical Response Time			
Mechanical Response Time	45 seconds		
CTS Bottle Concentration	Value: 99.57		
CTS Compound Concentration Avg	Value: 100.70		
Tolerance	5.00%		
Difference between measured and expected	1.13%		
System (Equilibration) Response Time Target (Spike)			
Equilibration Response Time	45 seconds		
Spike Reported	Value: 253.14		
Spike Expected	Value: 249.80		
System (Zero) Response Time			
System Zero Response Time	45 seconds		
System Response Time	45 seconds		

11.0 D6348 Annexes 1-8

D6348 Annexes 1. Test Plan Requirements**Annex 1.2**

The test quality objectives completed for the emissions test are demonstrated throughout Annexes 1, 2, 3, 4, 5, 6, 7 & 8 as laid out per ASTM D6348-03. All reference methods, pre-test and post test procedures were within acceptable limits. Data generated during the pre-test and post-test procedures are summarized below in order of the distinctive Annex.

Three 60 minute test runs were performed. The final analyte concentrations are the average of each test run. Data was taken at 60 second intervals. Each 60 second measurement was the average of 600 scans.

Acetaldehyde is used as the surrogate compound for the Annex 5 Spiking Technique due to Acetaldehyde being the closest spectral analysis match to Formaldehyde found in the combustion process of natural gas.

Annex Table 1.2.1 Certified Calibration Bottle Concentrations

Bottle	Expiration	NO2	Ethylene	SF6	Acetaldehyde	O2 (%)
CC476212	8/26/2025			10.19	100.3	
CC502013	2/12/2028	98.74				10.59%
EB0115896	12/18/2027		99.57			
CC440196	7/13/2032					20.92%

Cylinder # CC333629 Expiration: 10-18-2032

	Propane	CO	NO	SF6
Bottle Value	249.80	501.50	254.60	9.99
Analyzer System Response	253.14	510.59	251.89	9.73
Percent Difference	1.34%	1.81%	1.07%	2.57%

Annex Table 1.2.2 Measurement System Capabilities

Parameter Measured	Gas	Concentration (ppm)	Path Length	Equilibration Time	Dilution Factor	% Recovery
Path Length	Ethylene	101.086	5.076			
5-Gas Spike Direct	Propane	251.144				
	SF6	9.932				
	CO	503.972				
	NO	250.528				
	Acetaldehyde	99.725				
2-Gas Spike Direct	SF6	10.200				
	Ethylene	100.698		43 seconds		
5-Gas Analyzer Response	Propane	253.138				
	SF6	9.731				
	CO	510.586		17 seconds		
	NO	251.887				
2-Gas Analyzer Response	Acetaldehyde	97.753				
	SF6	10.155		36 seconds		
Analyte Spike Recovery	Acetaldehyde & SF6				9.70%	110.06%
					6.26%	123.17%
					4.62%	129.41%
System Zero	Nitrogen			39 seconds		
Post Spike System	Propane	252.231				
	CO	498.558				
	NO	249.775				
	SF6	9.692				

GAS

Annex 1.3

Annex Table 1.3.1 Test Specific Target Analytes and Data Quality Objectives

Compounds	Infrared Analysis Region (cm-1)	Expected Concentration Range	Measurement System Achievable Minimum Detectable Concentrations	Required Measurement System Accuracy and Precision for Test Application
CO	2000-2200	0-1200 ppm	0.16267 ppm	4 ppm
NO	1875-2138	0-1000 ppm	0.4007 ppm	2 ppm
NO ₂	2700-2950	0-100 ppm	0.4899 ppm	2 ppm
VOC	2600-3200	0-100 ppm	1.8520 ppm Total VOC's	1 ppm per VOC
	910-1150			
	2550-2950			
CH ₂ O	2550-2850	0-100 ppm	0.7878 ppm	1 ppm
Interfering Compounds	* CO is analyzed in a separate analysis region than CO ₂ and H ₂ O			
CO ₂	926-1150	0-10%	0%	n/a
Water Vapor	3200-3401	0-22%	0.20%	n/a

* VOCs compiled of Acetaldehyde, Ethylene, Hexane, and Propane.

Annex 1.4

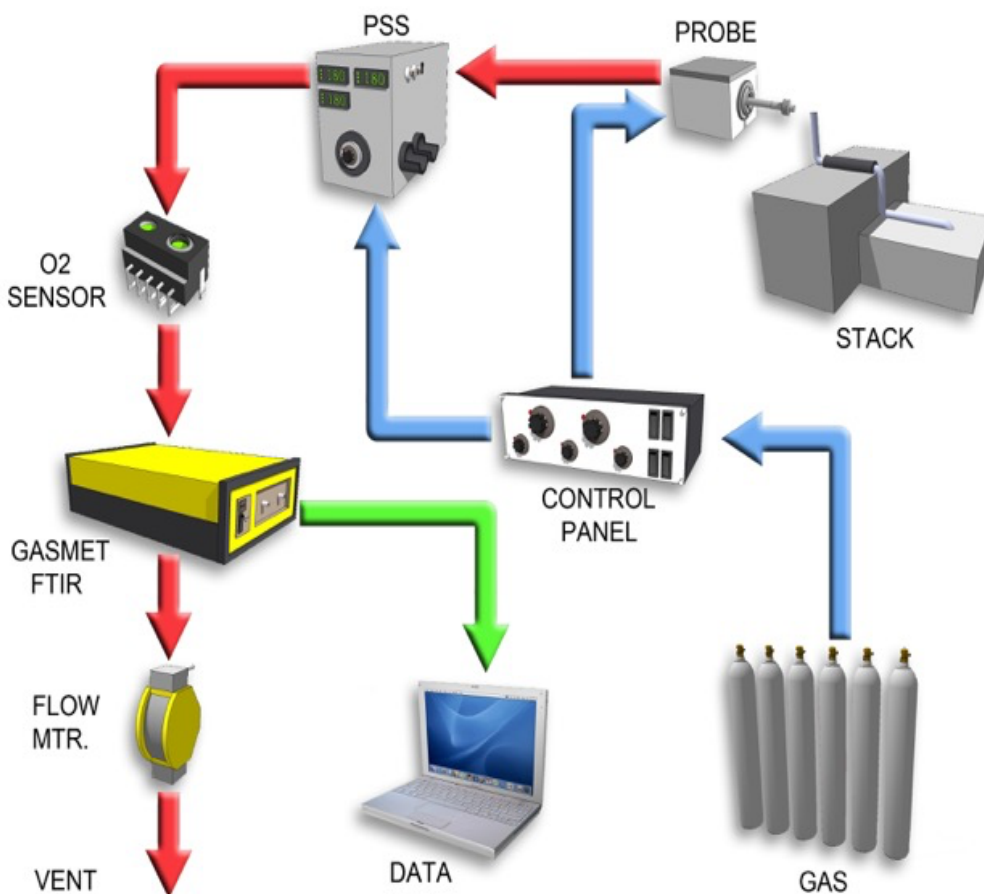
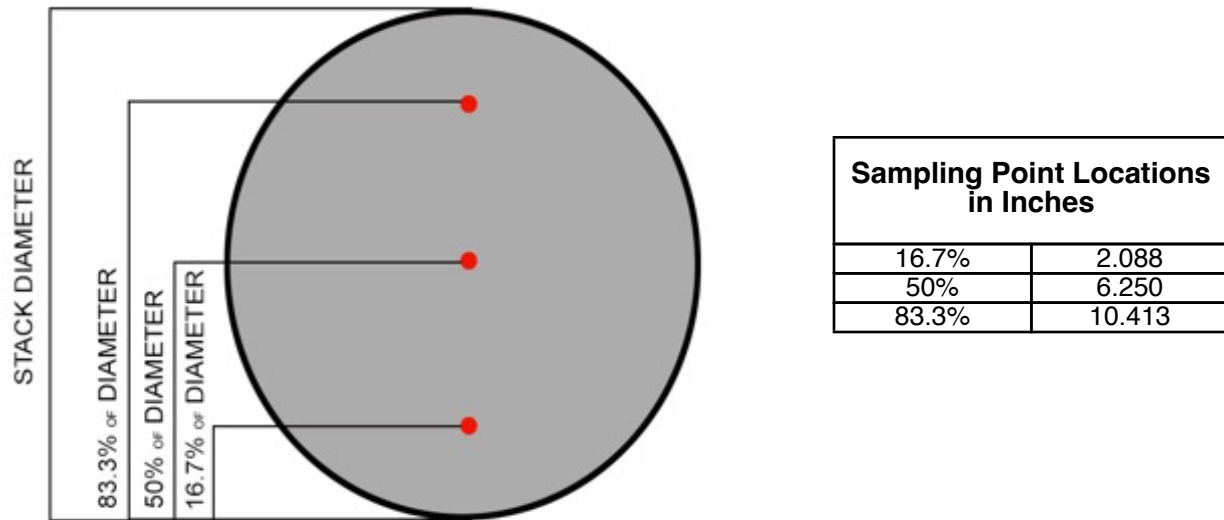


Figure Annex 1.4.1 Sampling Train

The testing instrumentation is housed in an enclosed vehicle which is located approximately 45 feet from the source. A heated sample line (sixty feet in length) is attached to the inlet of analyzer system and the source effluent discharges through the FTIR outlet.

GAS



TRI-PROBE SAMPLE POINT LOCATIONS AS PERCENTAGE OF STACK DIAMETER

Figure Annex 1.4.2 Sampling Points

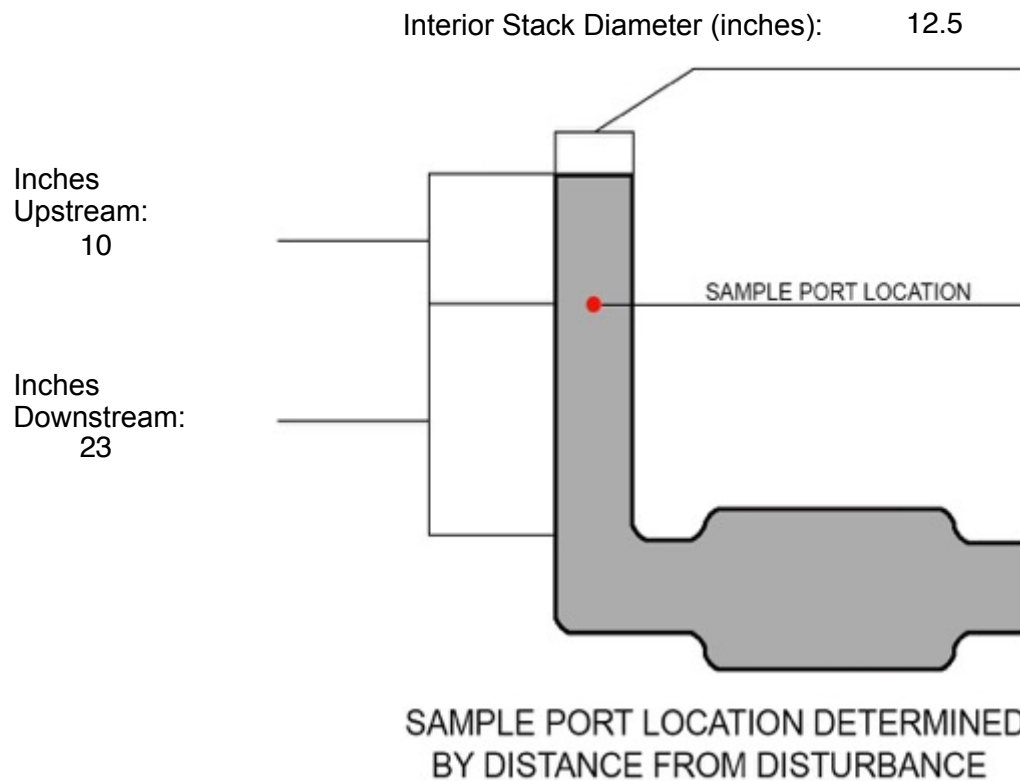


Figure Annex 1.4.3 Sample Port Location

Pressure @ Sampling Point (Pg "H ₂ O):	0.46
Temperature @ Sampling Point (Deg F):	833.21
Stack Gas Dry Volumetric Flow Rate	72394.07
H ₂ O%d @ Sampling Point:	18.18
CO ₂ %d @ Sampling Point:	11.32

D6348 Annex 2. Determination of FTIR Measurement System Minimum Detectable Concentrations (MDC#1)

Target Analyte	Results (ppm)
CO	0.1627
NO	0.4007
NO2	0.4899
Ethylene	0.3782
Propane	0.4351
Hexane	0.2233
Acetaldehyde	0.8152
Formaldehyde	0.7878

$$NEA_{rms}^m = \sqrt{\frac{1}{n} \sum_{j=1}^{N_m} (NEA_i^m)^2}$$

$$REF_{rms}^m = \sqrt{\frac{1}{n} \sum_{j=1}^{N_m} (REF_i^m)^2}$$

$$MDC\#1 = \frac{NEA_{rms}^m}{REF_{rms}^m} * \frac{C_{ref} L_{ref}}{L_{cell}}$$

D6348 Annex 3. FTIR Reference Spectra

Calibration Transfer Standard	Expected	Measured	Path Length	Validated
Ethylene	99.57	101.09	5.076	Passed

D6348 Annex 4. Required Pre-Test Procedures
Annex Table 1.2.2 Measurement System Capabilities

Parameter Measured	Gas	Concentration (ppm)	Path Length	Equilibration Time	Dilution Factor	% Recovery
Path Length	Ethylene	101.086	5.076			
5-Gas Spike Direct	Propane	251.144				
	SF6	9.932				
	CO	503.972				
	NO	250.528				
2-Gas Spike Direct	Acetaldehyde	99.725				
	SF6	10.200				
Mechanical Response Time	Ethylene	100.698		43 seconds		
5-Gas Analyzer Response	Propane	253.138		17 seconds		
	SF6	9.731				
	CO	510.586				
	NO	251.887				
2-Gas Analyzer Response	Acetaldehyde	0.000		36 seconds		
	SF6	10.155				
Analyte Spike Recovery	Acetaldehyde & SF6				9.70%	110.06%
					6.26%	123.17%
					4.62%	129.41%
System Zero	Nitrogen			39 seconds		
Post Spike System	Propane	252.231				
	CO	498.558				
	NO	249.775				
	SF6	9.692				

D6348 Annex 5. Analyte Spiking Technique

Parameter	Gas	Concentration	Measured	% Difference	Specification	Validated
Spike Direct	Propane	249.800	251.144	0.54%	+/- 2%	Pass
	SF6	9.988	9.932	0.56%	+/- 2%	Pass
	CO	501.500	503.972	0.49%	+/- 2%	Pass
	NO	254.600	250.528	1.60%	+/- 2%	Pass
	Acetaldehyde	100.300	99.725	0.57%	+/- 2%	Pass

Spike Run 1 via the System						
	Source Output	Spike Average	Dilution Factor	Expected	% Recovery	Specification
Acetaldehyde	1.256	11.956		10.863	110.060%	70-130%
SF6	0.000	0.989	9.958%			<10%

Spike Run 2 via the System						
	Source Output	Spike Average	Dilution Factor	Expected	% Recovery	Specification
Acetaldehyde	1.125	9.033		7.334	123.170%	70-130%
SF6	0.000	0.639	6.434%			<10%

Spike Run 3 via the System						
	Source Output	Spike Average	Dilution Factor	Expected	% Recovery	Specification
Acetaldehyde	1.145	7.410		5.726	129.410%	70-130%
SF6	0.000	0.471	4.742%			<10%

D6348 Annex 6. Determination of System Performance Parameters

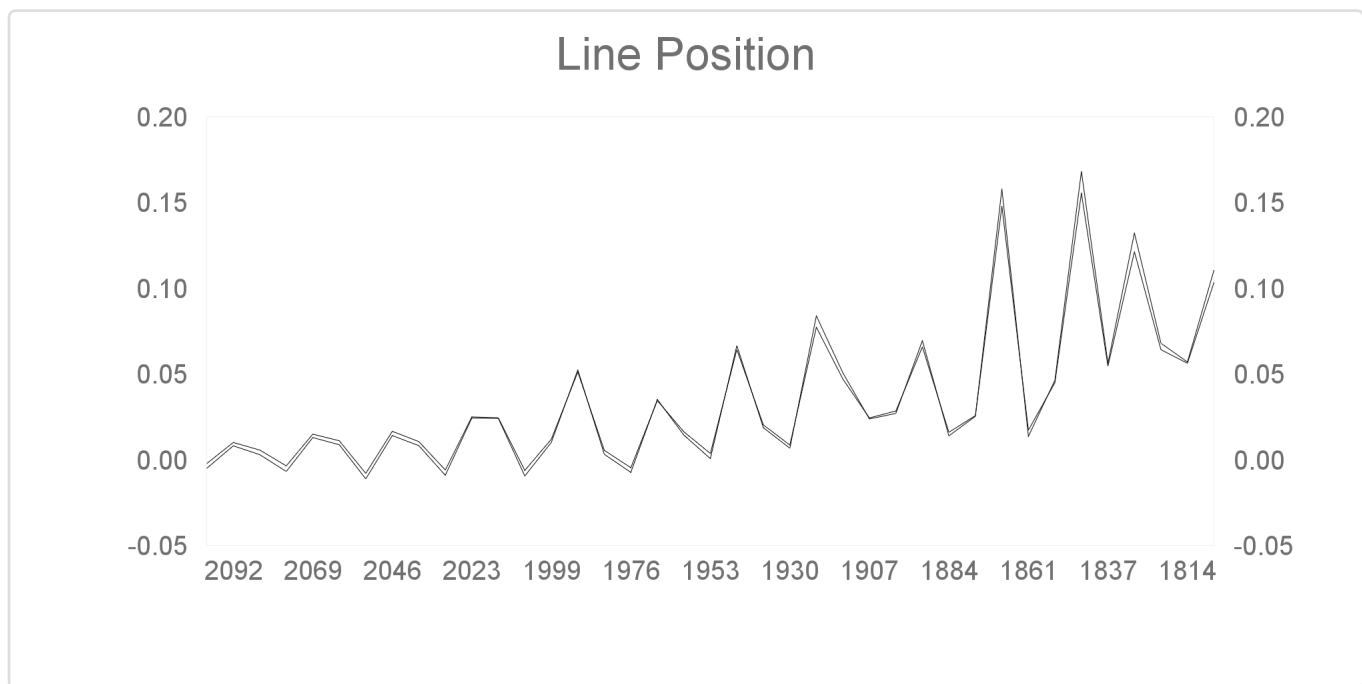
Noise Equivalent Absorbance (NEA)

RMS High 0.000340

RMS Mid 0.000199

RMS Low 0.000297

Line Position: Demonstrating the peak positions align correctly (Manual Comparison +/- 20%)



Line Position Difference: 0.00% Pass

Resolution Difference to Reference 0.53% Pass

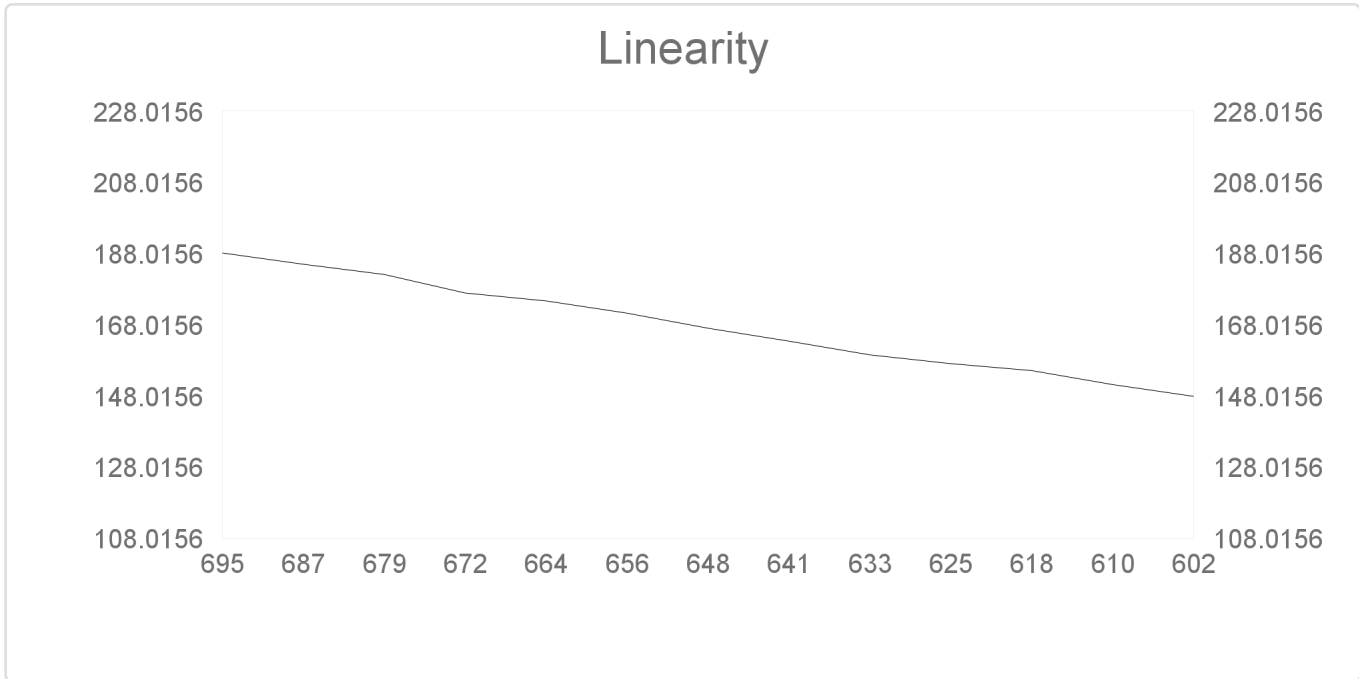
GAS

Resolution

The Gasmet GICCOR (Genzel Interferometer with Cube Corner Retroreflectors) interferometer is specially designed for maximum optical throughput and maximum signal to noise ratio of 7.72 (cm-1) remaining stable with any vibration and temperature changes.

Detector Linearity

The Gasmet DX4000 is a low resolution spectrometer where the aperture is fixed to a maximum angle setting and the detector linearity was tested with an alternate approach. A three point linearity of the CTS gas was performed and validated.



Regression Slope:

40.6540179 Pass

D6348 Annex 7. Preparation of Analytical Quantification Algorithm

The analytical accuracy of the quantification algorithm is satisfied via the results from Annex 5 per Annex 7.6

D6348 Annex 8. Post Test Quality Assurance/Quality Control Procedures

POST CTS System Check:	
CTS Bottle Concentration:	99.57
CTS Sample Concentration Average:	99.77
Difference between measured and expected:	0.20%
Tolerance:	5.00%

Run Data Validation - Automated vs Manual	Reading	Validated	Comments
Run 1 Points 1 & 2 on CO/NO/Propane	All within 20%	Passed	Demonstrates no interferences observed.
Run 2 Points 1 & 2 on CO/NO/Propane	All within 20%	Passed	Demonstrates no interferences observed.
Run 3 Points 1 & 2 on CO/NO/Propane	All within 20%	Passed	Demonstrates no interferences observed.

12.0 Signature Page

Job/File Name: Merit Energy Company; Hugoton Compressor Station; APC-164; ZZZZ



We certify that based on review of test data, knowledge of those individuals directly responsible for conducting this test, we believe the submitted information to be accurate and complete.

Company:	G.A.S. Inc.	Date: 5/15/2025
Print Name:	Travis Hartley	
Title:	Director of Stack Testing	
Signature:		
Phone Number:	580-225-0403	

Company:	G.A.S. Inc.	Date: 5/15/2025
Print Name:	Hunter Newton	
Title:	Emissions Specialist	

Company:		
Print Name:		Date:
Signature:		
Title:		
Phone Number:		

Appendices

Spike (5 Gas)



Airgas Specialty Gases
Airgas USA LLC
12722 S. Wentworth Ave.
Chicago, IL 60628
Airgas.com

CERTIFICATE OF ANALYSIS

Grade of Product: EPA PROTOCOL STANDARD

Part Number:	E05NI94E15AC014	Reference No:	54-403166330-1
Cylinder Number:	CC333629	Cylinder Volume:	147.0
Laboratory:	124 - Chicago (SAP) - IL	Cylinder Pressure:	2015
PGVP Number:	B12024	Valve Outlet:	660
Gas Code:	CO,CO2,NO,NOX,PPN,BALN	Certification Date:	Oct 18, 2024

Expiration Date: Oct 18, 2032

Certification performed in accordance with "EPA Traceability Protocol for Assay and Certification of Gaseous Calibration Standards (May 2012)" document EPA 600/R-12/531, using the assay procedures listed. Analytical Methodology does not require correction for analytical interference. This cylinder has a total analytical uncertainty as stated below with a confidence level of 95%. There are no significant impurities which affect the use of this calibration mixture. All concentrations are on a mole/mole basis unless otherwise noted. The results relate only to the items tested. The report shall not be reproduced except in full without approval of the laboratory. Do Not Use This Cylinder below 100 psig, i.e. 0.7 megapascals.

ANALYTICAL RESULTS

Component	Requested Concentration	Actual Concentration	Protocol Method	Total Relative Uncertainty	Absolute Uncertainty	Assay Dates
NOX	250.0 PPM	255.4 PPM	GI	+/- 1.4% NIST Traceable	+/- 3.6PPM NIST Traceable	10/18/2024 10/18/2024
NITRIC OXIDE	250.0 PPM	254.6 PPM	GI	+/- 1.4% NIST Traceable	+/- 3.6PPM NIST Traceable	10/18/2024 10/18/2024
PROPANE	250.0 PPM	249.8 PPM	GI	+/- 0.6% NIST Traceable	+/- 1.6PPM NIST Traceable	10/18/2024
CARBON MONOXIDE	500.0 PPM	501.5 PPM	GI	+/- 0.4% NIST Traceable	+/- 1.8PPM NIST Traceable	10/18/2024
CARBON DIOXIDE	5.000 %	4.980 %	GI	+/- 1.1% NIST Traceable	+/- 0.1% NIST Traceable	10/18/2024
NITROGEN	Balance					

CALIBRATION STANDARDS

Type	Lot ID	Cyl No	Concentration	Uncertainty	Exp Date
GMIS	DCK1128202232	CC754457	249.1 PPM NITRIC OXIDE NITROGEN	+/- 0.9%	Jan 18, 2031
RGM	12425	EB0112909	250.5 PPM NITRIC OXIDE NITROGEN	+/- 0.8%	Oct 19, 2026
PRIMARY	122-402992913-1	CC507051	4.22 PPM NITROGEN DIOXIDE AIR	+/- 10%	Mar 21, 2027
SRM	1680b	FF20776	494.8 PPM CARBON MONOXIDE NITROGEN	+/- 0.2%	Feb 07, 2028
NTRM	20060203	6162649Y	243.3 PPM PROPANE NITROGEN	+/- 0.5%	Mar 17, 2027
GMIS	160402954567106	CC301815	497.6 PPM CARBON MONOXIDE NITROGEN	+/- 0.3%	Apr 04, 2032
NTRM	13060426	CC413698	7.489 % CARBON DIOXIDE NITROGEN	+/- 0.6%	May 14, 2025

The SRM, NTRM, PRM, or RGM noted above is only in reference to the GMIS used in the assay and not part of the analysis.

ANALYTICAL EQUIPMENT

Instrument/Make/Model	Analytical Principle	Last Multipoint Calibration
Nicolet iS50 AUP2010242	FTIR	Oct 03, 2024
CO-1 SIEMENS ULTRAMAT 6E NIJ5700	NDIR	Oct 03, 2024
Nicolet iS50 AUP2010242	FTIR	Oct 03, 2024
Nicolet iS50 AUP2010242	FTIR	Oct 03, 2024
Nicolet iS50 AUP2110277	FTIR	Oct 03, 2024

NON-EPA ANALYTICAL RESULTS

Component	Requested/Max Concentration	Concentration
SULFUR HEXAFLUORIDE	10.00 PPM	9.988 PPM

Triad Data Available Upon Request



Signature on File

Approved for Release

9% O2/NO2



Airgas Specialty Gases
Airgas USA LLC
525 North Industrial Loop Road
Tooele, UT 84074
Airgas.com

CERTIFICATE OF ANALYSIS

Grade of Product: EPA PROTOCOL STANDARD

Part Number:	E03NI89E15W0020	Reference Number:	153-403252877-1
Cylinder Number:	CC502013	Cylinder Volume:	145.3 CF
Laboratory:	124 - Tooele (SAP) - UT	Cylinder Pressure:	2015 PSIG
PGVP Number:	B72025	Valve Outlet:	660
Gas Code:	NO2,O2,BALN	Certification Date:	Feb 12, 2025

Expiration Date: Feb 12, 2028

Certification performed in accordance with "EPA Traceability Protocol for Assay and Certification of Gaseous Calibration Standards (May 2012)" document EPA 600/R-12/531, using the assay procedures listed. Analytical Methodology does not require correction for analytical interference. This cylinder has a total analytical uncertainty as stated below with a confidence level of 95%. There are no significant impurities which affect the use of this calibration mixture. All concentrations are on a mole/mole basis unless otherwise noted. The results relate only to the items tested. The report shall not be reproduced except in full without approval of the laboratory. Do Not Use This Cylinder below 100 psig, i.e. 0.7 megapascals.

ANALYTICAL RESULTS					
Component	Requested Concentration	Actual Concentration	Protocol Method	Total Relative Uncertainty	Assay Dates
NITROGEN DIOXIDE	100.0 PPM	98.74 PPM	G1	+/- 2.0% NIST Traceable	02/05/2025, 02/12/2025
OXYGEN	10.60 %	10.59 %	G1	+/- 0.7% NIST Traceable	02/05/2025
NITROGEN	Balance				

CALIBRATION STANDARDS					
Type	Lot ID	Cylinder No	Concentration	Uncertainty	Expiration Date
GMIS	1534002024510	CC511408	59.65 PPM NITROGEN DIOXIDE/NITROGEN	1.8%	Apr 04, 2027
PRM	12438	D153564	59.5 PPM NITROGEN DIOXIDE/AIR	1.7%	Nov 21, 2024
NTRM	11060608	CC338459	14.93 % OXYGEN/NITROGEN	0.2%	Nov 02, 2028

The SRM, NTRM, PRM, or RGM noted above is only in reference to the GMIS used in the assay and not part of the analysis.

ANALYTICAL EQUIPMENT		
Instrument/Make/Model	Analytical Principle	Last Multipoint Calibration
MKS FTIR NO2 018143349	FTIR	Jan 16, 2025
Horiba MPA-510 W603MM58 O2	O2 Paramagnetic (DIXON)	Jan 09, 2025

Triad Data Available Upon Request



Signature on file
Approved for Release

Page 1 of 1

GAS

Ethylene Only



Airgas Specialty Gases
Airgas USA LLC
12722 S. Wentworth Ave.
Chicago, IL 60628
Airgas.com

CERTIFICATE OF ANALYSIS

Grade of Product: PRIMARY STANDARD

Part Number:	X02NI99P15ACVH8	Reference Number:	54-403224032-1
Cylinder Number:	EB0115896	Cylinder Volume:	144.0 CF
Laboratory:	124 - Chicago (SAP) - IL	Cylinder Pressure:	2015 PSIG
Analysis Date:	Dec 18, 2024	Valve Outlet:	350
Lot Number:	54-403224032-1		

Expiration Date: Dec 18, 2027

Primary Standard Gas Mixtures are traceable to N.I.S.T. weights and/or N.I.S.T. Gas Mixture reference materials.

ANALYTICAL RESULTS

Component	Req Conc	Actual Concentration (Mole %)	Analytical Uncertainty
ETHYLENE	100.0 PPM	99.57 PPM	+/- 1%
NITROGEN	Balance		



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Page 1 of 1

GAS



Airgas Mid South region
Airgas USA LLC
9741 E. 56th St. North
Tulsa, OK 74117
Airgas.com

CERTIFICATE OF ANALYSIS

Grade of Product: ULTRA HIGH PURITY-PURE

Part Number:	NI UHP300	Reference Number:	29-403219707-1
Cylinder Number:	4641953Y	Cylinder Volume:	304.0 CF
Laboratory:	106 - Tulsa Fast Fill (SAP) - OK	Cylinder Pressure:	2640 PSIG
Analysis Date:	Dec 13, 2024	Valve Outlet:	580
Lot Number:	29-403219707-1		

ANALYTICAL RESULTS

Component	Requested Purity	Certified Concentration
NITROGEN	99.999 %	99.999 %
THC	0.5 PPM	0.23 PPM
CO + CO2	1.0 PPM	0.1 PPM
Moisture	1.0 PPM	0.709 PPM
Oxygen	1.0 PPM	0.96 PPM

Impurities verified against analytical standards traceable to NIST by weight and/or analysis.



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Approved for Release

Page 1 of 1

21% O2



Airgas Specialty Gases
Airgas USA LLC
12722 S. Wentworth Ave.
Chicago, IL 60628
Airgas.com

CERTIFICATE OF ANALYSIS

Grade of Product: EPA PROTOCOL STANDARD

Part Number: E02NI79E15A00B1 Reference Number: 54-403116776-1
Cylinder Number: CC440196 Cylinder Volume: 146.0 CF
Laboratory: 124 - Chicago (SAP) - IL Cylinder Pressure: 2015 PSIG
PGVP Number: B12024 Valve Outlet: 590
Gas Code: O2,BALN Certification Date: Aug 13, 2024

Expiration Date: Aug 13, 2032

Certification performed in accordance with "EPA Traceability Protocol for Assay and Certification of Gaseous Calibration Standards (May 2012)" document EPA 600/R-12/531, using the assay procedures listed. Analytical Methodology does not require correction for analytical interference. This cylinder has a total analytical uncertainty as stated below with a confidence level of 95%. There are no significant impurities which affect the use of this calibration mixture. All concentrations are on a mole/mole basis unless otherwise noted. The results relate only to the items tested. The report shall not be reproduced except in full without approval of the laboratory. Do Not Use This Cylinder below 100 psig, i.e. 0.7 megapascals.

ANALYTICAL RESULTS					
Component	Requested Concentration	Actual Concentration	Protocol Method	Total Relative Uncertainty	Assay Dates
OXYGEN	21.00 %	20.92 %	G1	+/- 1.4% NIST Traceable	08/13/2024
NITROGEN	Balance				

CALIBRATION STANDARDS					
Type	Lot ID	Cylinder No	Concentration	Uncertainty	Expiration Date
NTRM	08010217	K011889	23.20 % OXYGEN/NITROGEN	+/- 0.4%	Feb 16, 2030

ANALYTICAL EQUIPMENT		
Instrument/Make/Model	Analytical Principle	Last Multipoint Calibration
O2-1 HORIBA MPA-510 W2TP7BN5	PARAMAGNETIC	Aug 07, 2024

Triad Data Available Upon Request



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Page 1 of 1

GAS

Acetaldehyde & SF6



Airgas Specialty Gases
Airgas USA LLC
9810 BAY AREA BLVD
Pasadena, TX 77507
Airgas.com

CERTIFICATE OF ANALYSIS

Grade of Product: PRIMARY STANDARD

Part Number:	X03NI99P15ACD42	Reference Number:	163-403122990-1
Cylinder Number:	CC476212	Cylinder Volume:	140.0 CF
Laboratory:	124 - Pasadena (SG06) - TX	Cylinder Pressure:	2015 PSIG
Analysis Date:	Aug 26, 2024	Valve Outlet:	350SS
Lot Number:	163-403122990-1		

Expiration Date: Aug 26, 2025

Primary Standard Gas Mixtures are traceable to N.I.S.T. weights and/or N.I.S.T. Gas Mixture reference materials.

ANALYTICAL RESULTS

Component	Req Conc	Actual Concentration (Mole %)	Analytical Uncertainty
SULFUR HEXAFLUORIDE	10.00 PPM	10.19 PPM	+/- 5%
ACETALDEHYDE	100.0 PPM	100.3 PPM	+/- 2%
NITROGEN	Balance		



Signature on file

Approved for Release

Page 1 of 1

GAS

Tri Probe Certification



Great Plains Analytical Services
303 W 3rd St
Elk City, OK, 73644
(580)225-0403 Fax: (580)225-2612

CERTIFICATE OF ANALYSIS

Grade of Product: CERTIFIED STANDARD-PROBE

Part Number:	12L	Reference Number:	25
Laboratory:	GAS INC.	Stack Diameter:	12"
Analysis Date:	6/14/2022	Target Flow Rate:	3L/min
LOT Number:	A	Number of Points:	3
SN:	25A 12L		

Product performance verified by direct comparison to calibration standards traceable to N.I.S.T.

*The probe listed on this form meets the multipoint traverse requirement of EPA Method 7e, section 8.4 as shown in the accompanying data. Method 7e, section 8.4 states that the multipoint traverse requirement can be satisfied by sampling via "a multi-hole probe designed to sample at the prescribed points with a flow +/- 10 percent of mean flow rate".

ANALYTICAL RESULTS

Run #	Total Flow (L/m)	Measured Flow Port A (L/m) (Delta q1)	Measured Flow Port B (L/m) (Delta q2)	Measured Flow Port C (L/m) (Delta q3)	Mean Probe Port Sampled Flow (L/m)
Run 1	2 LPM	0.710 (5.97△)	0.670 (0.00△)	0.630 (-5.97△)	0.670
Run 2	4 LPM	1.40 (5.79△)	1.32 (-0.25△)	1.25 (-5.54△)	1.32

*Calibration conducted in accordance with Emission Measurement Center Guideline Document – EMC GD-031

Notes:

Approved for Release

6/14/2022

Date

Probe size: 12L



UNITED STATES ENVIRONMENTAL PROTECTION AGENCY
RESEARCH TRIANGLE PARK, NC 27711

OFFICE OF
AIR QUALITY PLANNING
AND STANDARDS

March 15, 2021

Mr. Jordan Williamson
CEO
GAS Inc.
303 W. 3rd Street
Elk City, OK 73644

Dear Mr. Williamson:

We are writing in response to your letter received on September 17, 2020, in which you request the approval of alternative testing procedures. The EPA's Office of Air Quality Planning and Standards (OAQPS) is the delegated authority for consideration of major alternatives to test methods and procedures as set forth in 40 CFR parts 60 and 63 under which your request must be addressed. GAS Inc. is requesting a change to one of the test methods, ASTM D6348-03, used for conducting performance tests to determine compliance under 40 CFR part 60, Subpart JJJJ – Standards of Performance for Stationary Spark Ignition Internal Combustion Engines (Subpart JJJJ) and 40 CFR part 63, Subpart ZZZZ – National Emissions Standards for Hazardous Air Pollutants for Performance for Stationary Reciprocating Internal Combustion Engines (Subpart ZZZZ). The change being requested will be used to check detector linearity of the Fourier Transform Infrared (FTIR) instrumentation used to conduct this method. Specifically, you are requesting that the procedures of section 8.3.3 of Method 320 (40 CFR part 60, Appendix A), another FTIR-based method allowed under Subparts JJJJ and ZZZZ, be used in lieu of section A6.4.1 of ASTM D6348-03 when conducting testing using ASTM D6348-03 under 40 CFR part 60, Subpart JJJJ and 40 CFR part 63, Subpart ZZZZ.

In your request, you state that this alternative linearity check procedure will produce consistent results when utilizing either Method 320 or ASTM D6348-03. Additionally, some FTIR instrumentation does not allow for reducing the size of the aperture in the instrument and, thus, it would not be feasible to properly conduct the entirety of the ASTM D6348-03 method in its current form using such an instrument.

Based on our understanding of FTIR instrument principles and recognition that the requested alternative determination of detector linearity is both technically sound and contained within Method 320, we are approving the requested change. We believe that this alternative is acceptable for use in testing all engines subject to 40 CFR part 60 Subpart JJJJ and 40 CFR part 63, Subpart ZZZZ. Also, we will post this letter as ALT-141 on EPA's website (at www.epa.gov/emc/broadly-applicable-approved-alternative-testmethods) to announce that our approval of this alternative test method is broadly applicable to engines for the purposes of meeting Subparts JJJJ and ZZZZ.

If you should have any questions or require further information regarding this approval, please contact David Nash of my staff at 919-541-9425 or email at nash.dave@epa.gov.

Sincerely,

**STEFFAN
JOHNSON** Digitally signed by
STEFFAN JOHNSON
Date: 2021.03.15
12:28:50 -04'00'

Steffan M. Johnson, Group Leader
Measurement Technology Group

cc:

Sara Ayers, EPA/OECA/OC/MAMPD, (ayres.sara@epa.gov)
Melanie King, EPA/OAR/OAQPS/SPPD, (king.melanie@epa.gov)
James Leather, EPA Region 6, (leather.james@epa.gov)
David Nash, EPA/OAR/OAQPS/AQAD, (nash.dave@epa.gov)

Hunter Newton

GAS

580-225-0403

info@gasinc.us

Type of Sources Tested:

Stationary Internal Combustion Engines

- 4 Stroke Rich Burn Engines
- 2 Stroke & 4 Stroke Lean Burn Engines

Stationary Natural Gas Fired Generators

Stationary Propane Fired Generators

Gas Fired Boilers

Types of Analyzers:

- Gasmet DX4000 FTIR
- Gasmet Portable Sampling Unit with Zirconium Oxide O2 Sensory
- Testo 350
- Flame Ionization Detector

Qualifications:

Trained, studied, and fully demonstrates compliance for emissions testing via data collection outlined in the following Reference Methods:

- EPA Method 1 & 1A – Sampling & Traverse Points
- EPA Method 2 & 2C – Velocity & Volumetric Flow Rate of a Gas Stream
- EPA Method 3A – Oxygen
- EPA Method 7E – NOX
- EPA Method 10 – Carbon Monoxide
- EPA Method 25A – Volatile Organic Compounds
- ASTM D6348 – Extractive Fourier Transform Infrared Spectroscopy

Conducts emissions testing on a weekly basis including, but not limited to, the following test types: Initial Compliance, Biennial Compliance, Semiannual Compliance & Quarterly Compliance. All tests performed are in accordance to any and all Federal & State requirements as applicable (i.e. JJJJ, ZZZZ, 106.512, 117, PEA, etc.). Performed testing in Colorado, Utah, Wyoming, North Dakota, Montana, Kansas, New Mexico, Oklahoma, Texas, Louisiana (land and off-shore), Arkansas, Ohio, Pennsylvania, West Virginia, New York, Kentucky, & Mississippi.

- Quarterly Performance Reviews covering ongoing changes with Federal Regulations, State Compliance guidelines, & site-specific safety certifications.



RAW Result Data

1 Minute Average Data Sheet			
Client:	Merit Energy Company	Make:	Waukesha
Location:	Hugoton Compressor Station	Model:	L7042GSI
Unit Number:	APC-164	Serial Number:	PP168

Date	Time	H2O (%)	CO2 (%)	CO (ppmvw)	NO (ppmvw)	NO2 (ppmvw)	CH4 (ppmvw)	N2O (ppmvw)	NH3 (ppmvw)	SO2 (ppmvw)	C2H4 (ppmvw)	HCL (ppmvw)	HF (ppmvw)	C2H6 (ppmvw)	C3H8 (ppmvw)	C6H14 (ppmvw)	CHOH (ppmvw)	C2H4O (ppmvw)	SF6 (ppmvw)	Nox (ppmvw)	VOC (ppmvw)	Ambient Press. (mbar)	Oxygen (%)	Cell Temp (C).
Stability																								
5/15/2025	06:45	0.040	1.610	0.000	0.000	0.000	0.000	0.000	11.984	274.961	0.000	0.000	0.774	0.000	1.521	0.000	2.104	0.657	0.000	0.00	1.96	928.0	0.00	180.0
Pre Test Baseline Direct																								
5/15/2025	06:53	-0.000	0.010	0.032	0.000	0.000	0.055	0.001	0.033	0.000	0.042	0.136	0.000	0.024	0.016	0.005	0.049	0.000	0.002	0.00	0.05	927.0	0.00	180.0
5/15/2025	06:54	-0.000	0.002	0.083	0.296	0.000	0.000	0.009	0.000	0.000	0.002	0.000	0.012	0.000	0.000	0.005	0.000	0.035	0.000	0.30	0.03	927.0	0.00	180.0
5/15/2025	06:54	-0.002	0.001	0.045	0.000	0.072	0.000	0.005	0.004	0.025	0.000	0.063	0.000	0.000	0.000	0.001	0.000	0.000	0.001	0.07	0.00	928.0	0.00	180.0
CTS Direct																								
5/15/2025	06:59	0.011	0.027	0.000	0.913	0.000	0.000	0.000	0.132	0.000	101.370	0.027	0.000	0.238	0.000	0.017	0.004	0.096	0.018	0.91	67.68	926.0	0.00	180.0
5/15/2025	06:59	0.011	0.000	0.000	0.840	0.000	0.000	0.000	0.106	0.000	101.086	0.000	0.000	0.248	0.000	0.022	0.041	0.093	0.022	0.84	67.50	925.0	0.00	180.0
5/15/2025	07:00	0.010	0.000	0.000	1.009	0.000	0.000	0.006	0.279	0.000	100.801	0.000	0.026	0.151	0.000	0.015	0.027	0.000	0.021	1.01	67.23	926.0	0.00	180.0
High O2 Direct																								
5/15/2025	07:06	-0.000	0.006	0.000	0.000	0.182	0.159	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.009	0.030	0.000	0.18	0.02	925.0	20.82	180.0
5/15/2025	07:06	-0.001	0.007	0.000	0.000	0.075	0.030	0.000	0.000	0.000	0.000	0.090	0.000	0.000	0.000	0.005	0.000	0.031	0.000	0.08	0.03	923.0	20.79	180.0
5/15/2025	07:07	-0.002	0.002	0.000	0.000	0.150	0.000	0.000	0.005	0.000	0.000	0.022	0.019	0.000	0.000	0.006	0.000	0.100	0.003	0.15	0.08	922.0	20.78	180.0
NO2/Mid O2 Direct																								
5/15/2025	07:27	-0.007	0.000	0.000	0.000	98.287	0.489	0.340	0.000	0.000	0.000	0.000	0.000	0.017	0.000	0.000	0.025	0.000	0.000	98.29	0.00	914.0	10.52	180.0
5/15/2025	07:27	-0.008	0.003	0.000	0.000	98.277	0.584	0.333	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.017	0.000	0.000	98.28	0.00	917.0	10.52	180.0
5/15/2025	07:28	-0.009	0.004	0.027	0.000	98.726	0.660	0.340	0.000	0.000	0.000	0.057	0.000	0.228	0.000	0.000	0.009	0.000	0.000	98.73	0.00	916.0	10.52	180.0
Spike Direct																								
5/15/2025	07:34	-0.003	4.633	504.023	250.685	0.000	5.971	2.786	2.068	0.163	0.112	0.447	0.275	1.405	251.442	1.191	0.000	2.221	9.949	250.69	255.38	917.0	0.00	180.0
5/15/2025	07:34	-0.004	4.649	503.908	250.872	0.000	6.284	2.838	2.096	0.314	0.155	0.481	0.273	1.650	251.293	1.155	0.000	2.249	9.942	250.87	255.21	916.0	0.00	180.0
5/15/2025	07:35	-0.004	4.645	504.745	250.468	0.000	6.251	2.828	2.160	0.428	0.215	0.501	0.251	2.679	251.153	0.984	0.025	2.104	9.920	250.47	254.67	918.0	0.00	180.0
5/15/2025	07:35	-0.004	4.634	503.210	250.085	0.000	6.058	2.826	2.159	0.358	0.150	0.367	0.215	1.351	250.689	1.123	0.000	2.076	9.916	250.09	254.42	917.0	0.00	180.0

Date	Time	H2O (%)	CO2 (%)	CO (ppmvw)	NO (ppmvw)	NO2 (ppmvw)	CH4 (ppmvw)	N2O (ppmvw)	NH3 (ppmvw)	SO2 (ppmvw)	C2H4 (ppmvw)	HCL (ppmvw)	HF (ppmvw)	C2H6 (ppmvw)	C3H8 (ppmvw)	C6H14 (ppmvw)	CHOH (ppmvw)	C2H4O (ppmvw)	SF6 (ppmvw)	Nox (ppmvw)	VOC (ppmvw)	Ambient Press. (mbar)	Oxygen (%)	Cell Temp (C).
Run 1 - A 30minute/60min																								
5/15/2025	10:40	19.669	9.144	3527.211	87.556	7.703	281.126	6.792	1350.469	29.031	32.857	0.243	0.000	32.728	7.309	0.273	0.000	2.117	0.000	95.26	31.17	896.0	0.00	180.0
5/15/2025	10:41	19.022	9.281	3160.098	98.164	8.277	283.862	7.433	1337.048	26.923	32.421	0.312	0.000	33.151	7.308	0.242	0.000	1.212	0.000	106.44	30.22	896.0	0.00	180.0
5/15/2025	10:42	18.435	9.233	3492.856	95.776	7.980	284.338	6.981	1328.678	26.140	32.786	0.268	0.000	33.083	7.364	0.242	0.000	1.177	0.000	103.76	30.49	896.0	0.00	180.0
5/15/2025	10:43	18.189	9.265	3061.766	101.147	7.409	279.745	7.582	1304.841	25.690	31.373	0.339	0.000	32.944	7.321	0.267	0.000	2.021	0.000	108.56	30.12	896.0	0.00	179.0
5/15/2025	10:44	18.136	9.287	3546.750	96.722	7.370	279.484	7.126	1322.811	24.702	33.170	0.335	0.000	32.726	7.165	0.226	0.000	1.848	0.000	104.09	30.96	896.0	0.00	180.0
5/15/2025	10:45	18.079	9.269	3663.875	92.987	7.866	279.363	7.118	1328.183	23.980	33.857	0.311	0.000	32.956	7.156	0.212	0.000	1.828	0.000	100.85	31.37	896.0	0.00	180.0
5/15/2025	10:46	18.126	9.299	3400.807	97.273	7.201	279.134	7.524	1317.888	24.073	32.642	0.346	0.000	33.107	7.218	0.234	0.000	1.844	0.000	104.47	30.68	896.0	0.00	180.0
5/15/2025	10:47	18.126	9.258	3959.477	87.930	7.914	281.103	6.835	1331.456	24.348	34.138	0.324	0.000	33.221	7.254	0.185	0.000	1.876	0.000	95.84	31.63	896.0	0.00	180.0
5/15/2025	10:48	18.133	9.300	3570.460	88.591	7.899	280.693	7.018	1311.831	24.688	32.509	0.332	0.000	33.307	7.311	0.201	0.000	1.888	0.000	96.49	30.64	896.0	0.00	180.0
5/15/2025	10:49	18.144	9.278	3345.352	96.012	8.615	280.031	7.432	1321.867	24.766	33.058	0.427	0.000	33.255	7.536	0.176	0.000	1.061	0.000	104.63	30.63	896.0	0.00	180.0
5/15/2025	10:50	18.068	9.208	4047.763	86.399	8.459	282.030	6.545	1329.345	25.020	33.898	0.294	0.000	33.302	7.648	0.194	0.000	1.045	0.000	94.86	31.33	897.0	0.00	180.0
5/15/2025	10:51	18.121	9.264	3241.373	98.628	8.784	279.737	7.542	1320.005	24.970	32.495	0.294	0.000	33.011	7.833	0.197	0.000	1.140	0.000	107.41	30.65	896.0	0.00	180.0
5/15/2025	10:54	18.728	9.148	3647.460	97.363	8.529	279.111	7.185	1345.796	25.903	33.960	0.364	0.000	32.820	7.808	0.211	0.000	1.119	0.000	105.89	31.62	896.0	0.00	180.0
5/15/2025	10:55	18.115	9.215	3666.375	99.333	8.990	283.953	7.330	1319.614	25.263	32.885	0.335	0.000	33.040	7.855	0.191	0.000	1.080	0.000	108.32	30.88	896.0	0.00	180.0
5/15/2025	10:56	18.087	9.197	3686.147	97.300	8.695	283.953	7.118	1332.522	25.446	33.734	0.273	0.000	33.417	7.860	0.169	0.000	1.025	0.000	105.99	31.37	896.0	0.00	180.0
5/15/2025	10:57	18.043	9.213	3599.563	101.150	9.262	284.805	7.354	1323.421	25.730	32.993	0.374	0.000	33.373	7.859	0.175	0.000	1.122	0.000	110.41	30.95	896.0	0.00	180.0
5/15/2025	10:58	18.027	9.245	3229.700	110.259	9.633	285.650	7.905	1309.137	26.063	32.018	0.322	0.000	33.462	7.724	0.163	0.000	1.031	0.000	119.89	30.08	896.0	0.00	180.0
5/15/2025	10:59	18.050	9.218	3506.820	108.560	8.909	287.287	7.546	1311.936	26.200	32.429	0.328	0.000	33.766	7.656	0.168	0.000	1.095	0.000	117.47	30.34	896.0	0.00	180.0
5/15/2025	11:00	18.911	9.104	3712.774	101.719	8.742	284.388	7.277	1333.348	26.605	32.921	0.357	0.000	32.972	7.440	0.199	0.048	1.152	0.000	110.46	30.55	896.0	0.00	180.0
5/15/2025	11:01	17.988	9.246	3281.266	113.466	8.791	280.707	8.093	1310.774	25.654	31.586	0.286	0.000	32.501	7.289	0.178	0.000	1.017	0.000	122.26	29.38	896.0	0.00	180.0
5/15/2025	11:02	17.955	9.257	3655.712	108.560	8.541	283.725	7.651	1311.951	25.431	32.416	0.361	0.000	32.881	7.210	0.174	0.000	0.933	0.000	117.10	29.79	896.0	0.00	180.0
5/15/2025	11:03	17.969	9.256	3182.096	115.643	7.957	280.133	8.339	1301.882	25.529	31.184	0.308	0.000	32.341	7.139	0.179	0.000	0.937	0.000	123.60	28.91	896.0	0.00	180.0
5/15/2025	11:04	19.036	9.070	3678.589	113.357	8.654	280.624	7.817	1338.231	25.827	33.130	0.409	0.000	32.478	7.162	0.181	0.000	1.099	0.000	122.01	30.35	896.0	0.00	180.0
5/15/2025	11:05	18.200	9.217	3308.484	117.216	8.368	281.139	8.329	1311.377	25.946	31.700	0.320	0.000	32.624	7.156	0.168	0.000	1.014	0.000	125.58	29.30	896.0	0.00	180.0
5/15/2025	11:06	17.979	9.995	3535.830	116.864	8.607	282.518	8.002	1320.490	25.327	32.606	0.318	0.000	32.793	7.195	0.135	0.000	0.979	0.000	125.47	29.86	896.0	0.00	180.0
5/15/2025	11:07	18.759	9.113	3692.608	114.326	9.611	282.607	7.883	1315.746	26.425	31.873	0.318	0.000	32.660	7.131	0.139	0.011	1.088	0.000	123.94	29.38	896.0	0.00	180.0
5/15/2025	11:08	17.861	9.245	3507.802	112.906	9.021	284.518	7.921	1306.641	25.607	31.559	0.266	0.000	33.178	7.246	0.127	0.000	0.968	0.000	121.93	29.19	896.0	0.00	180.0
5/15/2025	11:09	17.843	9.977	3822.155	113.230	9.123	284.004	7.630	1313.479	25.508	32.282	0.312	0.000	32.941	7.183	0.125	0.000	0.929	0.000	122.35	29.57	896.0	0.00	180.0
5/15/2025	11:10	18.068	9.218	3668.788	110.299	9.273	282.664	7.796	1304.326	25.541	31.655	0.250	0.000	32.678	7.104	0.148	0.000	1.014	0.000	119.57	29.18	896.0	0.00	180.0
5/15/2025	11:11	18.463	9.132	3730.643	109.335	9.662	282.715	7.531	1334.314	25.967	32.622	0.303	0.000	32.869	7.013	0.129	0.000	1.017	0.000	119.00	29.70	896.0	0.00	180.0
Avg.		18.278	9.272	3537.687	102.936	8.528	282.172	7.488	1321.647	25.610	32.625	0.321	0.000	32.986	7.382	0.187	0.002	1.256	0.000	111.463	30.343	896.03	0.000	180.0

GAS

Date	Time	H2O (%)	CO2 (%)	CO (ppmvw)	NO (ppmvw)	NO2 (ppmvw)	CH4 (ppmvw)	N2O (ppmvw)	NH3 (ppmvw)	SO2 (ppmvw)	C2H4 (ppmvw)	HCL (ppmvw)	HF (ppmvw)	C2H6 (ppmvw)	C3H8 (ppmvw)	C6H14 (ppmvw)	CHOH (ppmvw)	C2H4O (ppmvw)	SF6 (ppmvw)	Nox (ppmvw)	VOC (ppmvw)	Ambient Press. (mbar)	Oxygen (%)	Cell Temp (C).
5/15/2025	13:28	17.983	9.171	4014.644	139.782	8.727	294.064	8.246	1348.439	25.459	32.024	0.401	0.000	34.992	7.223	0.284	0.000	0.957	0.000	148.51	29.78	895.0	0.00	180.0
5/15/2025	13:29	18.021	9.200	3808.380	136.565	8.379	294.493	8.296	1344.150	25.593	31.395	0.435	0.000	35.093	7.184	0.278	0.000	1.492	0.000	144.94	29.67	895.0	0.00	180.0
5/15/2025	13:30	18.022	9.222	3566.193	137.608	9.286	295.279	8.604	1329.283	25.596	30.098	0.390	0.000	35.314	7.145	0.258	0.000	0.964	0.000	146.89	28.37	895.0	0.00	180.0
5/15/2025	13:31	18.047	9.227	3647.743	140.682	9.118	293.728	8.645	1337.341	25.587	30.590	0.384	0.000	35.039	7.011	0.290	0.000	1.019	0.000	149.80	28.66	895.0	0.00	180.0
5/15/2025	13:32	18.094	9.204	3717.148	136.057	9.156	294.119	8.443	1336.466	25.572	30.561	0.387	0.000	34.901	6.950	0.249	0.000	0.970	0.000	145.21	28.47	895.0	0.00	180.0
5/15/2025	13:33	18.085	9.238	3484.875	142.569	9.453	294.500	8.996	1325.725	25.773	29.790	0.437	0.000	34.952	6.878	0.250	0.000	1.594	0.000	152.02	28.30	895.0	0.00	180.0
5/15/2025	13:34	18.048	9.175	4179.968	134.806	9.695	299.480	8.020	1340.785	25.886	31.847	0.332	0.000	35.653	6.926	0.245	0.000	1.024	0.000	144.50	29.33	895.0	0.00	180.0
5/15/2025	13:35	18.048	9.171	4123.449	122.256	9.699	298.888	7.629	1340.460	25.839	30.852	0.337	0.000	35.936	6.948	0.286	0.000	1.013	0.000	131.96	28.76	895.0	0.00	180.0
5/15/2025	13:36	18.087	9.203	3751.557	130.281	9.607	294.263	8.151	1344.557	25.719	30.884	0.396	0.000	35.195	6.811	0.254	0.000	0.973	0.000	139.89	28.56	894.0	0.00	180.0
5/15/2025	13:37	18.096	9.181	3803.176	131.551	9.045	296.334	8.171	1351.514	25.639	31.132	0.385	0.000	35.486	6.932	0.239	0.000	1.587	0.000	140.60	29.22	894.0	0.00	180.0

Avg.	18.106	9.187	3771.049	137.805	9.332	296.387	8.230	1340.490	25.922	31.462	0.367	0.000	35.482	7.099	0.256	0.000	1.145	0.000	147.137	29.349	894.90	0.000	180.0
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Spike 3 - 2-Gas

5/15/2025	13:45	16.989	8.473	3271.909	123.670	5.151	276.270	7.601	1259.339	25.156	24.173	0.431	0.000	31.102	6.887	0.474	0.000	7.410	0.471	128.82	28.89	894.0	0.00	180.0
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Run 3 - B 30minute/60min

5/15/2025	13:48	18.037	9.103	4620.336	114.576	9.323	297.347	6.928	1377.462	25.418	33.712	0.365	0.000	35.903	7.961	0.262	0.000	1.021	0.000	123.90	31.64	894.0	0.00	180.0
5/15/2025	13:49	18.088	9.159	4065.223	115.242	9.581	297.083	7.242	1363.318	25.832	31.734	0.416	0.000	36.061	8.048	0.296	0.000	0.943	0.000	124.82	30.42	894.0	0.00	180.0
5/15/2025	13:50	18.140	9.187	4020.760	113.957	9.524	298.889	7.286	1360.566	26.093	31.762	0.385	0.000	36.436	8.186	0.304	0.000	0.953	0.000	123.48	30.60	894.0	0.00	180.0
5/15/2025	13:51	18.137	9.193	3547.933	123.749	9.968	297.792	7.899	1352.050	26.347	30.548	0.245	0.000	36.263	8.263	0.326	0.000	1.097	0.000	133.72	30.01	894.0	0.00	180.0
5/15/2025	13:52	18.166	9.189	3639.411	127.142	9.593	296.367	7.966	1351.573	25.963	30.806	0.362	0.000	36.466	8.398	0.357	0.000	0.947	0.000	136.74	30.28	894.0	0.00	180.0
5/15/2025	13:53	18.144	9.213	3564.062	129.152	9.707	296.154	8.175	1348.210	35.051	30.542	0.364	0.000	36.369	8.428	0.291	0.000	1.026	0.000	138.86	30.06	894.0	0.00	180.0
5/15/2025	13:54	18.166	9.165	4022.303	126.395	9.575	298.444	7.723	1373.964	35.013	32.829	0.403	0.000	36.749	8.649	0.333	0.000	1.074	0.000	135.97	31.92	894.0	0.00	180.0
5/15/2025	13:55	18.141	9.193	4166.978	121.708	9.259	297.913	7.492	1369.553	35.341	32.896	0.315	0.000	36.541	8.725	0.330	0.000	0.857	0.000	130.97	31.89	894.0	0.00	180.0
5/15/2025	13:56	18.085	9.205	3941.278	124.646	8.782	297.993	7.847	1354.691	35.434	32.189	0.273	0.000	36.652	8.767	0.323	0.000	1.651	0.000	133.43	31.97	894.0	0.00	180.0
5/15/2025	13:57	18.014	9.125	3787.021	73.465	9.246	290.561	5.320	1448.410	24.730	30.957	0.252	0.000	35.119	7.933	0.359	0.000	1.386	0.000	82.71	30.21	894.0	0.00	180.0
5/15/2025	13:58	18.102	9.184	3360.641	87.423	9.040	292.231	7.169	1397.925	35.085	32.073	0.377	0.000	35.598	8.691	0.374	0.000	1.069	0.000	96.46	31.54	894.0	0.00	180.0
5/15/2025	13:59	18.112	9.214	3436.677	124.679	8.801	293.586	8.083	1366.239	34.775	32.181	0.328	0.000	35.841	8.970	0.362	0.000	0.932	0.000	133.48	31.77	894.0	0.00	180.0
5/15/2025	14:00	18.071	9.184	3806.257	124.728	8.382	294.473	7.538	1360.585	34.935	31.385	0.468	0.000	35.831	8.894	0.380	0.000	0.840	0.000	133.11	31.14	894.0	0.00	180.0
5/15/2025	14:01	18.067	9.165	4349.918	103.038	8.902	298.983	6.832	1366.605	35.212	32.965	0.312	0.000	36.840	9.139	0.334	0.000	0.924	0.000	111.94	32.40	894.0	0.00	180.0
5/15/2025	14:02	18.079	9.221	3515.178	127.774	8.589	295.845	8.044	1351.889	35.030	31.353	0.315	0.000	36.493	9.097	0.369	0.000	1.037	0.000	136.36	31.43	894.0	0.00	180.0
5/15/2025	14:03	18.096	9.254	3441.048	138.354	9.140	297.739	8.462	1332.506	34.783	30.666	0.419	0.000	36.975	9.101	0.394	0.000	1.059	0.000	147.49	31.04	894.0	0.00	180.0
5/15/2025	14:04	18.119	9.239	3525.927	140.391	8.493	298.024	8.398	1333.907	35.078	30.795	0.367	0.000	37.124	9.100	0.403	0.000	0.956	0.000	148.88	31.07	894.0	0.00	180.0

Date	Time	H2O (%)	CO2 (%)	CO (ppmvw)	NO (ppmvw)	NO2 (ppmvw)	CH4 (ppmvw)	N2O (ppmvw)	NH3 (ppmvw)	SO2 (ppmvw)	C2H4 (ppmvw)	HCL (ppmvw)	HF (ppmvw)	C2H6 (ppmvw)	C3H8 (ppmvw)	C6H14 (ppmvw)	CHOH (ppmvw)	C2H4O (ppmvw)	SF6 (ppmvw)	Nox (ppmvw)	VOC (ppmvw)	Ambient Press. (mbar)	Oxygen (%)	Cell Temp (C).
5/15/2025	14:40	0.005	0.036	0.000	0.000	0.303	0.000	0.021	1.841	0.000	0.093	0.000	0.011	0.000	0.000	0.000	0.000	99.365	10.181	0.30	66.31	896.0	0.00	180.0
5/15/2025	14:40	-0.000	0.033	0.000	0.000	0.335	0.005	0.000	1.790	0.000	0.057	0.000	0.000	0.230	0.000	0.000	0.000	99.725	10.189	0.33	66.53	896.0	0.00	180.0
5/15/2025	14:40	0.003	0.000	0.000	0.000	0.439	0.049	0.038	2.130	0.000	0.076	0.000	0.000	0.038	0.000	0.000	0.000	99.459	10.160	0.44	66.36	896.0	0.00	180.0

Post 5-Gas																								
5/15/2025	14:43	0.004	4.633	499.932	249.865	0.000	5.323	2.760	2.509	2.892	0.293	0.179	0.054	1.698	252.726	0.971	0.000	2.417	9.693	249.87	256.47	897.0	0.00	180.0
5/15/2025	14:43	-0.000	4.621	497.354	249.367	0.000	5.859	2.701	2.753	1.860	0.160	0.432	0.106	1.070	251.834	1.224	0.000	2.599	9.721	249.37	256.12	897.0	0.00	180.0
5/15/2025	14:44	0.005	4.628	498.327	249.657	0.000	5.807	2.796	2.655	1.730	0.146	0.307	0.136	0.391	252.051	1.369	0.000	2.669	9.700	249.66	256.67	897.0	0.00	180.0
5/15/2025	14:44	-0.000	4.632	498.822	249.546	0.000	6.061	2.754	2.702	1.487	0.186	0.244	0.036	1.451	252.330	1.145	0.000	2.835	9.710	249.55	256.63	897.0	0.00	180.0
5/15/2025	14:44	-0.000	4.630	499.097	249.726	0.000	5.671	2.753	2.207	2.329	0.260	0.208	0.078	1.785	252.318	1.105	0.000	2.652	9.700	249.73	256.47	897.0	0.00	180.0
5/15/2025	14:44	0.005	4.623	498.602	250.095	0.000	5.726	2.718	2.623	2.588	0.216	0.152	0.094	1.569	251.902	1.151	0.000	2.857	9.692	250.09	256.25	898.0	0.00	180.0
5/15/2025	14:44	0.001	4.632	497.759	249.421	0.000	5.391	2.754	2.508	1.922	0.225	0.083	0.057	1.316	252.061	1.088	0.000	2.364	9.678	249.42	255.96	898.0	0.00	180.0
5/15/2025	14:44	-0.000	4.647	498.979	250.091	0.000	5.647	2.738	2.430	2.097	0.342	0.336	0.012	2.498	252.518	0.929	0.000	2.843	9.662	250.09	256.50	898.0	0.00	180.0
5/15/2025	14:44	0.002	4.645	498.150	250.208	0.000	5.781	2.744	2.487	2.528	0.354	0.506	0.087	2.377	252.335	1.046	0.000	2.588	9.669	250.21	256.39	898.0	0.00	180.0

End of Test Purge																								
5/15/2025	14:48	-0.000	0.000	0.322	0.000	0.000	0.156	0.030	0.400	0.000	0.000	0.246	0.037	0.221	0.133	0.000	0.000	0.206	0.000	0.00	0.27	900.0	0.00	180.0
5/15/2025	14:49	-0.000	0.001	0.367	0.000	0.450	0.113	0.018	0.404	0.000	0.000	0.000	0.000	0.006	0.111	0.005	0.062	0.182	0.007	0.45	0.24	900.0	0.00	180.0
5/15/2025	14:49	-0.000	0.005	0.302	0.089	0.000	0.000	0.000	0.539	0.000	0.000	0.000	0.000	0.230	0.086	0.000	0.112	0.177	0.009	0.09	0.20	900.0	0.00	180.0
5/15/2025	14:49	-0.000	0.005	0.315	0.000	0.402	0.000	0.000	0.246	0.000	0.058	0.000	0.009	0.199	0.120	0.014	0.021	0.082	0.000	0.40	0.24	900.0	0.00	180.0
5/15/2025	14:49	-0.000	0.001	0.324	0.000	0.053	0.000	0.008	0.399	0.000	0.088	0.000	0.000	0.021	0.026	0.027	0.025	0.000	0.002	0.05	0.14	900.0	0.00	180.0
5/15/2025	14:49	-0.000	0.000	0.173	0.000	0.036	0.000	0.023	0.343	0.000	0.110	0.000	0.021	0.033	0.007	0.051	0.000	0.035	0.004	0.04	0.21	900.0	0.00	180.0
5/15/2025	14:49	-0.000	0.001	0.190	0.000	0.000	0.000	0.008	0.403	0.000	0.000	0.097	0.014	0.140	0.098	0.013	0.106	0.230	0.002	0.00	0.28	900.0	0.00	180.0
5/15/2025	14:49	-0.000	0.003	0.267	0.000	0.033	0.000	0.009	0.414	0.000	0.000	0.175	0.000	0.138	0.084	0.030	0.032	0.000	0.003	0.03	0.14	900.0	0.00	180.0
5/15/2025	14:49	-0.000	0.000	0.302	0.000	0.000	0.000	0.011	0.491	0.000	0.000	0.110	0.022	0.150	0.047	0.040	0.117	0.069	0.000	0.00	0.17	900.0	0.00	180.0
5/15/2025	14:49	-0.000	0.006	0.261	0.000	0.000	0.000	0.012	0.343	0.037	0.000	0.079	0.002	0.000	0.015	0.002	0.081	0.000	0.004	0.00	0.02	900.0	0.00	180.0